Sales Analyzer

Reading the file

```
file = open("sales_data.csv", "r")
lines = file.readlines()
print(lines)
```

```
['OrderID,Date,Customer,Product,Category,Quantity,Price\n',
'1001,2023-02-17,Rohit Gupta,Chair,Furniture,2,1999\n', '1002,2023-03-
04,Amit Sharma,Marker,Stationery,4,35\n', '1003,2023-08-31,Rahul
Mehta,Headphones,Electronics,2,1299\n', '1004,2023-08-27,Rohit
Gupta,Desk,Furniture,4,3499\n', '1005,2023-12-22,Rahul
Mehta,Headphones,Electronics,4,1299\n', '1006,2023-04-12,Rahul
Mehta,Marker,Stationery,1,35\n', '1007,2023-04-05,Amit
Sharma,Notebook,Stationery,3,99\n', '1008,2023-04-23,Amit
Sharma,Marker,Stationery,1,35\n', '1009,2023-07-09,Amit
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Sharma,Smartphone,Electronics,1,15999\n', '1011,2023-01-04,Priya
Singh,Mouse,Electronics,5,499\n', '1012,2023-07-30,Rohit
Gupta,Smartphone,Electronics,5,15999\n', '1013,2023-08-14,Rohit
Gupta,Laptop,Electronics,2,45999\n', '1014,2023-05-29,Rohit
Gupta,Notebook,Stationery,2,99\n', '1015,2023-11-25,Priya
Singh,Marker,Stationery,3,35\n', '1016,2023-03-04,Priya
Singh,Marker,Stationery,4,35\n', '1017,2023-09-05,Neha
Verma,Headphones,Electronics,5,1299\n', '1018,2023-03-16,Amit
Sharma,Laptop,Electronics,3,45999\n', '1019,2023-10-16,Rahul
Mehta,Printer,Electronics,2,8999\n', '1020,2023-01-17,Amit
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Singh,Smartphone,Electronics,2,15999\n', '1023,2023-04-20,Neha
Verma,Desk,Furniture,3,3499\n', '1024,2023-08-02,Amit
Sharma,Pen,Stationery,5,20\n', '1025,2023-09-20,Neha
Verma,Notebook,Stationery,3,99\n', '1026,2023-05-15,Rohit
Gupta,Notebook,Stationery,5,99\n', '1027,2023-10-23,Rahul
Mehta,Desk,Furniture,2,3499\n', '1028,2023-12-21,Anjali
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Mehta,Laptop,Electronics,5,45999\n', '1030,2023-08-31,Neha
Verma,Laptop,Electronics,3,45999\n', '1031,2023-07-02,Rohit
Gupta,Pen,Stationery,1,20\n', '1032,2023-10-01,Rohit
Gupta,Mouse,Electronics,2,499\n', '1033,2023-05-23,Priya
Singh,Headphones,Electronics,4,1299\n', '1034,2023-09-15,Rahul
Mehta,Notebook,Stationery,4,99\n', '1035,2023-08-26,Priya
Singh,Smartphone,Electronics,4,15999\n', '1036,2023-12-14,Amit
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Verma,Notebook,Stationery,1,99\n', '1038,2023-01-27,Neha
Verma,Marker,Stationery,1,35\n', '1039,2023-03-16,Amit
```

```
Verma,Marker,Stationery,1,35\n', '1972,2023-10-08,Rahul
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Mehta,Chair,Furniture,5,1999\n', '1975,2023-12-30,Neha
Verma,Headphones,Electronics,2,1299\n', '1976,2023-10-24,Anjali
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Chauhan,Headphones,Electronics,5,1299\n', '1979,2023-07-04,Rohit
Gupta,Chair,Furniture,2,1999\n', '1980,2023-10-24,Amit
Sharma,Mouse,Electronics,5,499\n', '1981,2023-05-23,Amit
Sharma,Mouse,Electronics,5,499\n', '1982,2023-04-23,Priya
Singh,Desk,Furniture,5,3499\n', '1983,2023-12-10,Anjali
Chauhan,Marker,Stationery,3,35\n', '1984,2023-09-30,Anjali
Chauhan,Laptop,Electronics,2,45999\n', '1985,2023-07-05,Neha
Verma,Printer,Electronics,4,8999\n', '1986,2023-07-22,Rahul
Mehta,Desk,Furniture,2,3499\n', '1987,2023-09-07,Amit
Sharma,Chair,Furniture,5,1999\n', '1988,2023-05-29,Neha
Verma,Marker,Stationery,3,35\n', '1989,2023-05-26,Rahul
Mehta,Printer,Electronics,1,8999\n', '1990,2023-10-29,Rahul
Mehta,Printer,Electronics,4,8999\n', '1991,2023-12-20,Anjali
Chauhan,Pen,Stationery,2,20\n', '1992,2023-03-23,Rohit
Gupta,Mouse,Electronics,3,499\n', '1993,2023-08-14,Anjali
Chauhan,Smartphone,Electronics,5,15999\n', '1994,2023-09-12,Rahul
Mehta,Laptop,Electronics,5,45999\n', '1995,2023-08-13,Priya
Singh,Chair,Furniture,2,1999\n', '1996,2023-02-05,Rahul
Mehta,Chair,Furniture,4,1999\n', '1997,2023-07-04,Rohit
Gupta,Pen,Stationery,2,20\n', '1998,2023-10-01,Rahul
Mehta,Headphones,Electronics,2,1299\n', '1999,2023-07-29,Priya
Singh,Notebook,Stationery,5,99\n', '2000,2023-08-31,Rohit
Gupta,Laptop,Electronics,4,45999\n']
```

```
# removing the 1st row as it is the heading section (String format)
# strip = used to remove the blank space basically /n
# split = used to seperate those different rows using ","  or else it
will all be merged as one line
# changing 5th and 6th column to an int value
# store the data in "field" within Newdata using append function


NewData = []
for line in lines[1: ]:
    field = line.strip().split(",")
    field[5] = int(field[5])
    field[6] = int(field[6])
    NewData.append(field)

print(NewData)
```

```
[['1001', '2023-02-17', 'Rohit Gupta', 'Chair', 'Furniture', 2, 1999],
['1002', '2023-03-04', 'Amit Sharma', 'Marker', 'Stationery', 4, 35],
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'Headphones', 'Electronics', 4, 1299], ['1963', '2023-03-08', 'Priya
Singh', 'Pen', 'Stationery', 5, 20], ['1964', '2023-06-27', 'Priya
Singh', 'Chair', 'Furniture', 4, 1999], ['1965', '2023-07-27', 'Rohit
Gupta', 'Mouse', 'Electronics', 4, 499], ['1966', '2023-03-27', 'Amit
Sharma', 'Headphones', 'Electronics', 2, 1299], ['1967', '2023-01-08',
'Rahul Mehta', 'Headphones', 'Electronics', 2, 1299], ['1968', '2023-
03-18', 'Amit Sharma', 'Headphones', 'Electronics', 2, 1299], ['1969',
'2023-01-24', 'Amit Sharma', 'Printer', 'Electronics', 1, 8999],
['1970', '2023-11-09', 'Priya Singh', 'Notebook', 'Stationery', 1,
99], ['1971', '2023-10-25', 'Neha Verma', 'Marker', 'Stationery', 1,
35], ['1972', '2023-10-08', 'Rahul Mehta', 'Smartphone',
'Electronics', 5, 15999], ['1973', '2023-03-11', 'Rohit Gupta',
'Chair', 'Furniture', 2, 1999], ['1974', '2023-11-11', 'Rahul Mehta',
'Chair', 'Furniture', 5, 1999], ['1975', '2023-12-30', 'Neha Verma',
'Headphones', 'Electronics', 2, 1299], ['1976', '2023-10-24', 'Anjali
Chauhan', 'Notebook', 'Stationery', 4, 99], ['1977', '2023-06-12',
'Amit Sharma', 'Headphones', 'Electronics', 4, 1299], ['1978', '2023-
12-27', 'Anjali Chauhan', 'Headphones', 'Electronics', 5, 1299],
['1979', '2023-07-04', 'Rohit Gupta', 'Chair', 'Furniture', 2, 1999],
['1980', '2023-10-24', 'Amit Sharma', 'Mouse', 'Electronics', 5, 499],
['1981', '2023-05-23', 'Amit Sharma', 'Mouse', 'Electronics', 5, 499],
['1982', '2023-04-23', 'Priya Singh', 'Desk', 'Furniture', 5, 3499],
['1983', '2023-12-10', 'Anjali Chauhan', 'Marker', 'Stationery', 3,
35], ['1984', '2023-09-30', 'Anjali Chauhan', 'Laptop', 'Electronics',
2, 45999], ['1985', '2023-07-05', 'Neha Verma', 'Printer',
'Electronics', 4, 8999], ['1986', '2023-07-22', 'Rahul Mehta', 'Desk',
'Furniture', 2, 3499], ['1987', '2023-09-07', 'Amit Sharma', 'Chair',
'Furniture', 5, 1999], ['1988', '2023-05-29', 'Neha Verma', 'Marker',
'Stationery', 3, 35], ['1989', '2023-05-26', 'Rahul Mehta', 'Printer',
'Electronics', 1, 8999], ['1990', '2023-10-29', 'Rahul Mehta',
'Printer', 'Electronics', 4, 8999], ['1991', '2023-12-20', 'Anjali
Chauhan', 'Pen', 'Stationery', 2, 20], ['1992', '2023-03-23', 'Rohit
Gupta', 'Mouse', 'Electronics', 3, 499], ['1993', '2023-08-14',
'Anjali Chauhan', 'Smartphone', 'Electronics', 5, 15999], ['1994',
'2023-09-12', 'Rahul Mehta', 'Laptop', 'Electronics', 5, 45999],
['1995', '2023-08-13', 'Priya Singh', 'Chair', 'Furniture', 2, 1999],
['1996', '2023-02-05', 'Rahul Mehta', 'Chair', 'Furniture', 4, 1999],
['1997', '2023-07-04', 'Rohit Gupta', 'Pen', 'Stationery', 2, 20],
['1998', '2023-10-01', 'Rahul Mehta', 'Headphones', 'Electronics', 2,
1299], ['1999', '2023-07-29', 'Priya Singh', 'Notebook', 'Stationery',
5, 99], ['2000', '2023-08-31', 'Rohit Gupta', 'Laptop', 'Electronics',
4, 45999]]
```

Q1) Total Revenue

```
total_revenue = 0
```

```
for row in NewData:
    quantity = row[5]
    price = row[6]
    total_revenue += quantity * price

print("Total Revenue: ₹", total_revenue)
```

```
Total Revenue: ₹ 26662151
```

Q2) Total number of orders

```
total_orders = len(NewData)
print("Total number of orders: " , total_orders)
```

```
Total number of orders:  1000
```

Q3) Total items(Quantity) were sold?

```
total_quantity = 0
for row in NewData:
    total_quantity += row[5]

print("Total Quantity: ", total_quantity)
```

```
Total Quantity:  3129
```

Q4) Total Revenue generated by each category

```
# categories = furniter, stationery, Electronics

category_revenue = {}

for row in NewData:
    category = row[4]
    revenue = row[5]*row[6]

    if category in category_revenue:
        category_revenue[category] += revenue
    else:
        category_revenue[category] = revenue

# print("Revenue by Category", category_revenue)

for cat, rev in category_revenue.items():  #items:retrieves a dynamic
view of all 'key-value' entries
    print(f"{cat}: ₹{rev}")
```

```
Furniture: ₹1702880
Stationery: ₹50660
Electronics: ₹24908611
```

Q5) Total revenue generated by each product?

```
product_revenue = {}

for row in NewData:
    product = row[3]
    revenue = row[5]*row[6]

    if product in product_revenue:
        product_revenue[product] += revenue
    else:
        product_revenue[product] = revenue

# print("Revenue by Product", product_revenue)

for prod, rev in product_revenue.items():  #items:retrieves a dynamic
view of all 'key-value' entries
    print(f"{prod}: ₹{rev}")
```

```
Chair: ₹621689
Marker: ₹9835
Headphones: ₹420876
Desk: ₹1081191
Notebook: ₹35145
Smartphone: ₹5487657
Mouse: ₹129740
Laptop: ₹16053651
Printer: ₹2816687
Pen: ₹5680
```

Q6-A) which product has the highest quantity sold?

```
product_quantity = {}

for row in NewData:
    product = row[3]
    quantity = row[5]

    if product in product_quantity:
        product_quantity[product] += quantity
    else:
        product_quantity[product] = quantity

for prod, qty in product_quantity.items():
    print(f"{prod}: {qty}")
```

```
Chair: 311
Marker: 281
Headphones: 324
Desk: 309
Notebook: 355
```

```
Smartphone: 343
Mouse: 260
Laptop: 349
Printer: 313
Pen: 284
```

Q6-B) which product has the highest quantity sold?

```
product_quantity = {}

for row in NewData:
    product = row[3]
    quantity = row[5]

    if product in product_quantity:
        product_quantity[product] += quantity
    else:
        product_quantity[product] = quantity

highest_Sold = None
max_quantity = 0

for product, qty in product_quantity.items():
    if qty > max_quantity:
        max_quantity = qty
        highest_Sold = product

print(f"Product with highest quantity sold: {highest_Sold} with
{max_quantity} units sold")
```

```
Product with highest quantity sold: Notebook with 355 units sold
```

Q7-A) Customers expenditure

```
customer_revenue = {}

for row in NewData:
    customer = row[2]
    revenue = row[5]*row[6]

    if customer in customer_revenue:
        customer_revenue[customer] += revenue
    else:
        customer_revenue[customer] = revenue

for cust, rev in customer_revenue.items():
    print(f"{cust}: ₹{rev}")
```

```
Rohit Gupta: ₹3284600
Amit Sharma: ₹3157156
```

```
Rahul Mehta: ₹5306609
Priya Singh: ₹5367141
Neha Verma: ₹5346104
Anjali Chauhan: ₹4200541
```

Q7-B) Which customer spent the most money?

```
customer_revenue = {}
for row in NewData:
    customer = row[2]
    revenue = row[5]*row[6]

    if customer in customer_revenue:
        customer_revenue[customer] += revenue
    else:
        customer_revenue[customer] = revenue

highest_revenue_customer = None
max_revenue = 0

for cust, rev in customer_revenue.items():
    if rev > max_revenue:
        max_revenue = rev
        highest_revenue_customer = cust

print(f"Customer with highest expenditure: {highest_revenue_customer}
with ₹{max_revenue}")
```

```
Customer with highest expenditure: Priya Singh with ₹5367141
```

Q8) Total Revenue For each date

```
daily_revenue = {}
for row in NewData:
    date = row[1]
    revenue = row[5]*row[6]

    if date in daily_revenue:
        daily_revenue[date] += revenue
    else:
        daily_revenue[date] = revenue

for date, rev in daily_revenue.items():
    print(f"{date}: ₹{rev}")
```

```
2023-02-17: ₹189992
2023-03-04: ₹93328
2023-08-31: ₹356857
2023-08-27: ₹14194
2023-12-22: ₹35287
```

```
2023-03-23: ₹1996
2023-09-25: ₹4293
2023-05-07: ₹6670
2023-07-08: ₹595
2023-01-28: ₹68029
2023-03-27: ₹6495
2023-04-24: ₹230993
2023-03-28: ₹92078
2023-04-03: ₹10462
2023-12-06: ₹91998
2023-11-24: ₹39893
2023-05-04: ₹15999
2023-02-07: ₹175
2023-01-09: ₹13996
2023-10-24: ₹48890
2023-05-12: ₹3998
2023-11-09: ₹17594
2023-11-12: ₹7996
2023-07-15: ₹228991
2023-05-24: ₹3977
2023-01-26: ₹60
2023-05-13: ₹46099
2023-06-25: ₹198
2023-12-03: ₹3957
2023-05-11: ₹35
2023-05-22: ₹7567
2023-07-27: ₹7192
2023-09-24: ₹100
2023-12-01: ₹3897
2023-01-24: ₹9139
2023-04-14: ₹80
2023-02-08: ₹79995
2023-06-28: ₹20
2023-09-26: ₹15999
2023-08-03: ₹495
2023-11-06: ₹6998
2023-10-17: ₹2059
2023-07-07: ₹1497
2023-09-13: ₹998
2023-04-21: ₹5196
2023-07-22: ₹6998
```

Q9) Total Revenue between two given dates?

```python
from datetime import datetime

start = input("Enter start date (YYYY-MM-DD): ")
end = input("Enter end date (YYYY-MM-DD): ")
```

```
start_date = datetime.strptime(start, "%Y-%m-%d")
end_date = datetime.strptime(end, "%Y-%m-%d")

total_dateRevenue = 0
for row in NewData:
    order_date = datetime.strptime(row[1], "%Y-%m-%d")
    if start_date <= order_date <= end_date:
        total_dateRevenue += row[5]*row[6]

print(f"Total revenue from {start} to {end}: ₹{total_dateRevenue}")
```

```
Total revenue from 2023-01-01 to 2023-01-31: ₹1454449
```

Q10) Export all orders under the "Electronics" category into a new CSV file

```
elecFile = open("electrical_sales.csv", "w")

elecFile.write("Order ID,Order Date,Customer Name,Product
Name,Category,Quantity Ordered,Price Each\n")
for row in NewData:
    if row[4] == "Electronics":
        elecFile.write(",".join(map(str, row)) + "\n")
elecFile.close()

print("All Electrical sales data saved to electrical_sales.csv")
```

```
All Electrical sales data saved to electrical_sales.csv
```

```
# Alternate method

furnitureFile_1 = open("furniture_sales.csv", "w")

furnitureFile_1.write("Order ID,Order Date,Customer Name,Product
Name,Category,Quantity Ordered,Price Each\n")
for row in NewData:
    if row[4] == "Furniture":
        row_str = ",".join([str(item) for item in row])
        furnitureFile_1.write(row_str + "\n")
furnitureFile_1.close()

print("All Furniture sales data saved to furniture_sales.csv")
```

```
All Furniture sales data saved to furniture_sales.csv
```